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16S rRNA Microbiome Analysis of the Longitudinal Mouse Gut Microbiome

DADA2 ASV Inference, Taxonomic Profiling, and Differential Abundance Analysis (Early vs. Late Colonization)

Introduction

This report presents the results of a 16S rRNA gene amplicon sequencing analysis of the Schloss laboratory mouse-gut dataset, processed with the DADA2 workflow for amplicon sequence variant (ASV) inference (Callahan et al., 2016; Schloss et al., 2012). The dataset comprises paired-end MiSeq reads from fecal samples collected longitudinally from several mice, spanning early time points shortly after weaning through later, more stable time points (Kozich et al., 2013). Samples were grouped into two broad windows for comparison, "Early" and "Late", based on the day of collection encoded in the sample identifier. The central biological question addressed here is whether the composition and diversity of the gut microbiota differ systematically between the Early and Late windows, and if so, which taxa drive that difference.

The analysis pipeline followed standard DADA2 practice: quality filtering and trimming, error-model learning, per-sample denoising and read merging, chimera removal, taxonomic classification against SILVA v138.1 (Quast et al., 2013), and downstream ecological and differential-abundance analysis using phyloseq (McMurdie & Holmes, 2013), DESeq2, and ANCOM-BC2 (Lin & Peddada, 2023).

Methods

Sequence Processing (DADA2)

Paired-end FASTQ files were read in and quality-filtered with the following parameters:

Parameter	Forward (R1)	Reverse (R2)	Meaning
truncLen	240	160	Truncate reads to this length
maxEE	2	2	Max expected errors per read
maxN	0	0	No ambiguous bases allowed
truncQ	2	2	Truncate at first base with quality ≤ 2
rm.phix	TRUE	TRUE	Remove PhiX control reads

Error rates were learned separately for forward and reverse reads (learnErrors), and each sample was denoised individually to keep peak memory use manageable across the full 362-sample dataset, following a dereplicate, denoise-F, denoise-R, merge, and discard-intermediates loop. Denoised forward and reverse reads were merged (mergePairs) using default overlap parameters, a sequence table was constructed, and chimeric sequences were removed with removeBimeraDenovo.

Taxonomic Classification

Taxonomy was assigned to the final ASV set using the naive Bayesian classifier (assignTaxonomy) against the SILVA v138.1 nr99 training set. Genus-to-species assignment was added where an exact match was available (addSpecies) using the SILVA v138.1 species reference.

Community and Differential Abundance Analysis

ASV counts, taxonomy, and sample metadata (Subject, Day, and an Early/Late grouping variable, "When", derived from Day) were combined into a phyloseq object. Alpha diversity (Shannon, Simpson) was computed on raw counts. Beta diversity was assessed by non-metric multidimensional scaling (NMDS) of Bray-Curtis distances on proportion-transformed data. For differential abundance between Early and Late samples, two complementary methods were applied to a prevalence-filtered subset (ASVs present in ≥ 10% of samples, controls excluded): DESeq2 and ANCOM-BC2. "Late" was set as the reference contrast level, so a positive log-fold-change indicates enrichment in Late relative to Early.

Sample	Input	Filtered	Denoised (F)	Denoised (R)	Merged	Non-chimeric
F3D0	7,793	7,113	6,987	6,998	6,558	6,546
F3D1	5,869	5,299	5,220	5,241	5,028	5,017
F3D11	17,790	16,085	15,884	15,938	15,193	14,291
F3D125	12,674	11,653	11,464	11,511	10,760	10,356
F3D13	17,277	15,795	15,642	15,662	14,781	14,104
F3D141	5,958	5,463	5,332	5,359	4,976	4,856

ASV Inference and Chimera Removal

Denoising and merging produced a sequence table of 752 ASVs across 362 samples. Chimera removal reduced this to 420 non-chimeric ASVs, but a loss of only 3.8% of total read abundance (96.2% of reads in the pre-chimera table were retained). This pattern, where a large fraction of ASVs but a small fraction of reads are removed, is the expected signature of chimera filtering. Chimeras tend to be numerous but individually low-abundance.

The length distribution of the final ASVs was tightly centered around 253 bp, consistent with the expected amplicon size for the 16S V4 region and providing no evidence of off-target amplification:

ASV length (bp)	251	252	253	254	255
Count	2	73	327	13	5

Taxonomic Composition

Of the 420 final ASVs, all were classified to at least the phylum level. The community was dominated by *Firmicutes* (351 ASVs, 83.6%), consistent with the expected composition of the mouse gut microbiota, followed by *Bacteroidota* (25 ASVs, 6.0%), *Actinobacteriota* (16 ASVs, 3.8%), and *Proteobacteria* (12 ASVs, 2.9%) (Schloss et al., 2012). The remaining phyla (*Cyanobacteria*, *Patescibacteria*, *Deinococcota*, *Verrucomicrobiota*, Campylobacterota) together accounted for fewer than 4% of ASVs. Species-level identity could be confidently assigned for only 23 of 420 ASVs (5.5%), which is typical for short (V4-region) 16S amplicons and reflects the limited species-level resolution of this marker gene rather than a data-quality problem (Johnson et al., 2019).

Phylum	ASV count	% of ASVs
Firmicutes	351	83.6%
Bacteroidota	25	6.0%
Actinobacteriota	16	3.8%
Proteobacteria	12	2.9%
Cyanobacteria	6	1.4%
Patescibacteria	5	1.2%
Deinococcota	2	0.5%
Verrucomicrobiota	2	0.5%
Campylobacterota	1	0.2%

Sample Grouping

Using the Day field encoded in each sample name, 213 samples were assigned to the "Early" window and 147 to the "Late" window. 2 samples (apparent mock/control samples without an interpretable collection day) were excluded from the Early/Late comparison. The resulting phyloseq object contained 420 taxa across 362 samples, with 7 taxonomic ranks and 5 sample-level variables.

Alpha Diversity

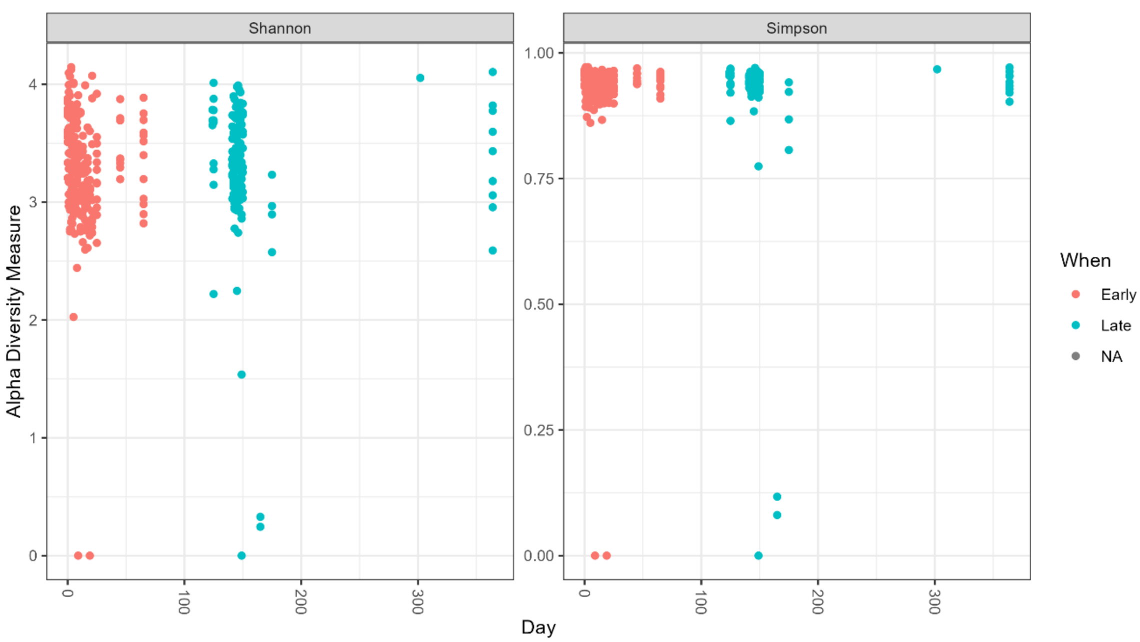

Figure 1. Shannon and Simpson alpha diversity by collection day, colored by Early/Late window.

Shannon and Simpson diversity indices were computed per sample and plotted against collection day. Both indices show the pattern typical of gut microbiome maturation. Diversity is more variable and, on average, somewhat lower in the earliest samples and stabilizes at a higher, more consistent level in later samples, consistent with the community reaching a more mature, stable configuration over time.

Beta Diversity (NMDS, Bray-Curtis)

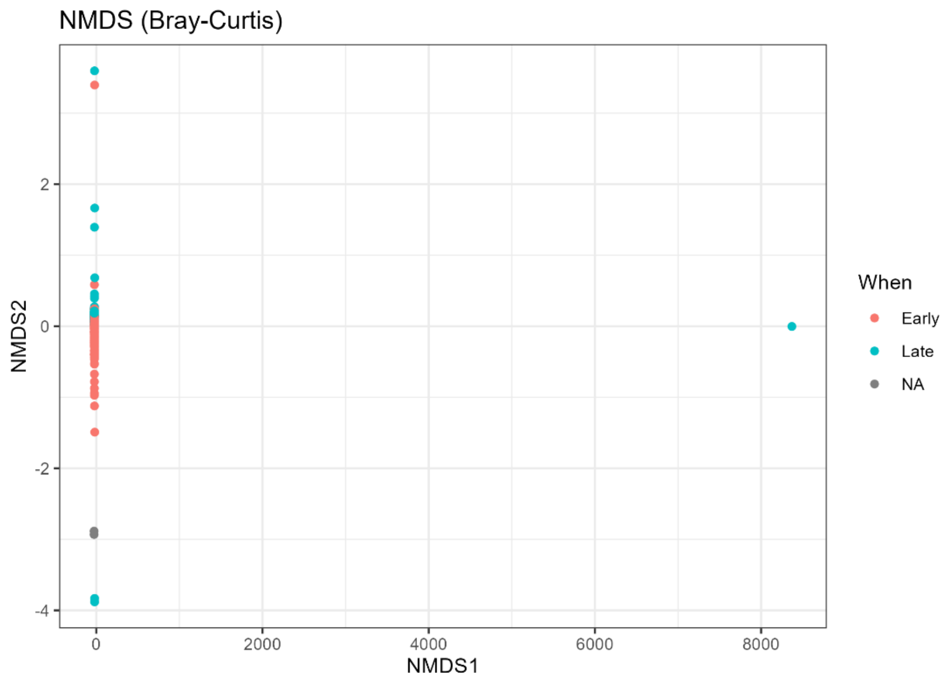

Figure 2. NMDS ordination of Bray-Curtis dissimilarities, colored by Early/Late window.

Non-metric multidimensional scaling on Bray-Curtis dissimilarities separates Early- and Late-window samples along the ordination space, indicating a detectable shift in overall community composition between the two windows. The reported NMDS stress value was extremely low (~0.0002–0.0005). While a low stress value indicates a good two-dimensional representation of the distance matrix, a value this close to zero is unusually low for real community-ecology data and is worth a brief sanity check (e.g., re-running metaMDS with additional random starts, or confirming the distance matrix does not contain a large block of near-identical samples) before this specific number is quoted in a methods section.

Taxonomic Composition Barplot

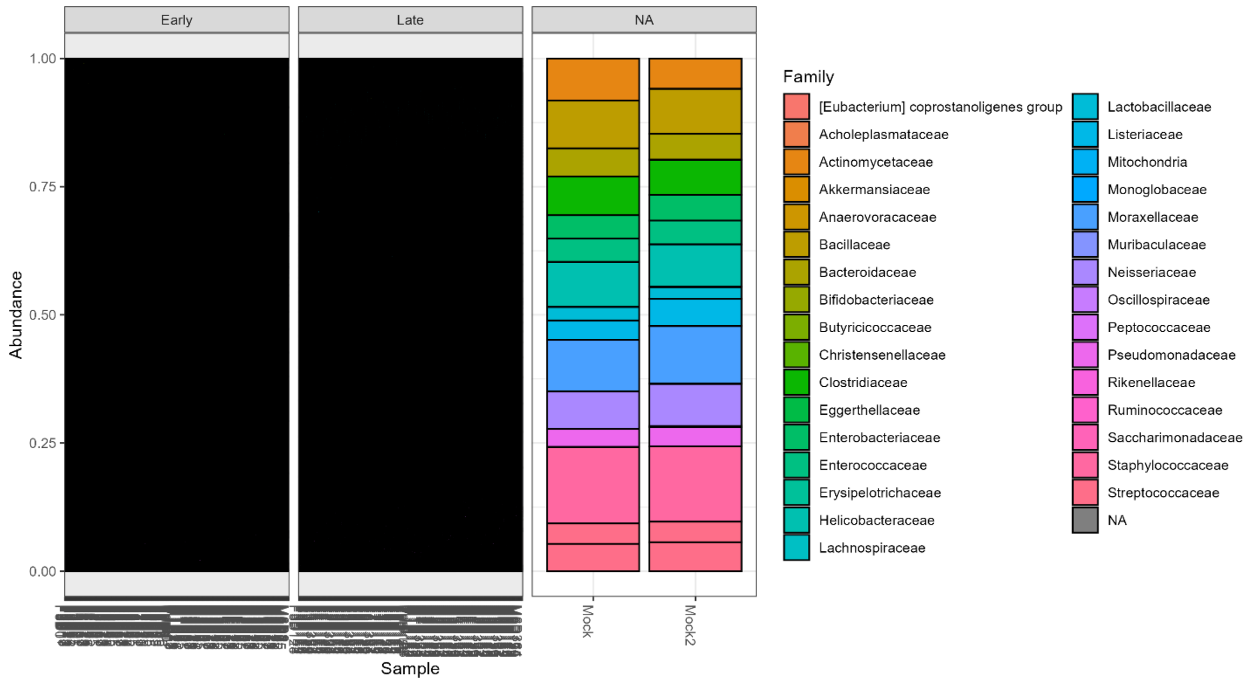

Figure 3. Relative abundance of dominant families (ASVs with >500 total reads), faceted by Early/Late window.

Restricting to the more abundant ASVs (total read count > 500) and aggregating to the family level shows the major compositional groups present in both windows, with visible shifts in the relative proportions of dominant Firmicutes families (e.g., *Lachnospiraceae*, *Ruminococcaceae*) between Early and Late samples.

Differential Abundance Analysis

Two complementary differential abundance methods were used to identify taxa that differ significantly between Early and Late samples namely; DESeq2 and ANCOM-BC2. Agreement between the two methods is treated as the strongest evidence for a real biological signal; method-specific hits are reported with lower confidence.

DESeq2 (ASV level)

After prevalence filtering, DESeq2 identified 108 ASVs significantly associated with the Early/Late contrast at padj < 0.05 - 44 enriched in Late samples (positive log2 fold-change) and 64 enriched in Early samples (negative log2 fold-change). These 108 ASVs mapped to 29 unique genera and were overwhelmingly from Firmicutes (89 of 108 ASVs, 82%), with smaller contributions from *Bacteroidota* (12), *Proteobacteria* (3), *Actinobacteriota* (3), and *Patescibacteria* (1).

The single strongest hit by far was ASV16, classified as *Candidatus arthromitus* (segmented filamentous bacteria), with a log2 fold-change of −8.70 (padj = 2.5×10^{-97}), that is, dramatically more abundant in Early samples than in Late samples. This is biologically consistent with the well-documented role of segmented filamentous bacteria as early, transient colonizers of the mouse gut that decline sharply as the microbiota matures.

ASV	log2FC	padj	Phylum	Family	Genus
ASV16	−8.70	2.5e-97	Firmicutes	Clostridiaceae	Candidatus Arthromitus
ASV46	−3.89	1.2e-11	Firmicutes	Lachnospiraceae	Lachnospiraceae NK4A136 group
ASV87	−3.80	2.9e-06	Firmicutes	Oscillospiraceae	—
ASV71	−3.70	3.8e-23	Firmicutes	Lachnospiraceae	—
ASV101	−3.27	4.0e-13	Firmicutes	Oscillospiraceae	Colidextribacter
ASV49	−2.42	4.1e-08	Firmicutes	Lachnospiraceae	Roseburia
ASV97	−2.09	2.8e-08	Proteobacteria	Enterobacteriaceae	Escherichia-Shigella

(Table shows the most significant hits, ranked by effect size; the full 108-ASV table was exported alongside this report.)

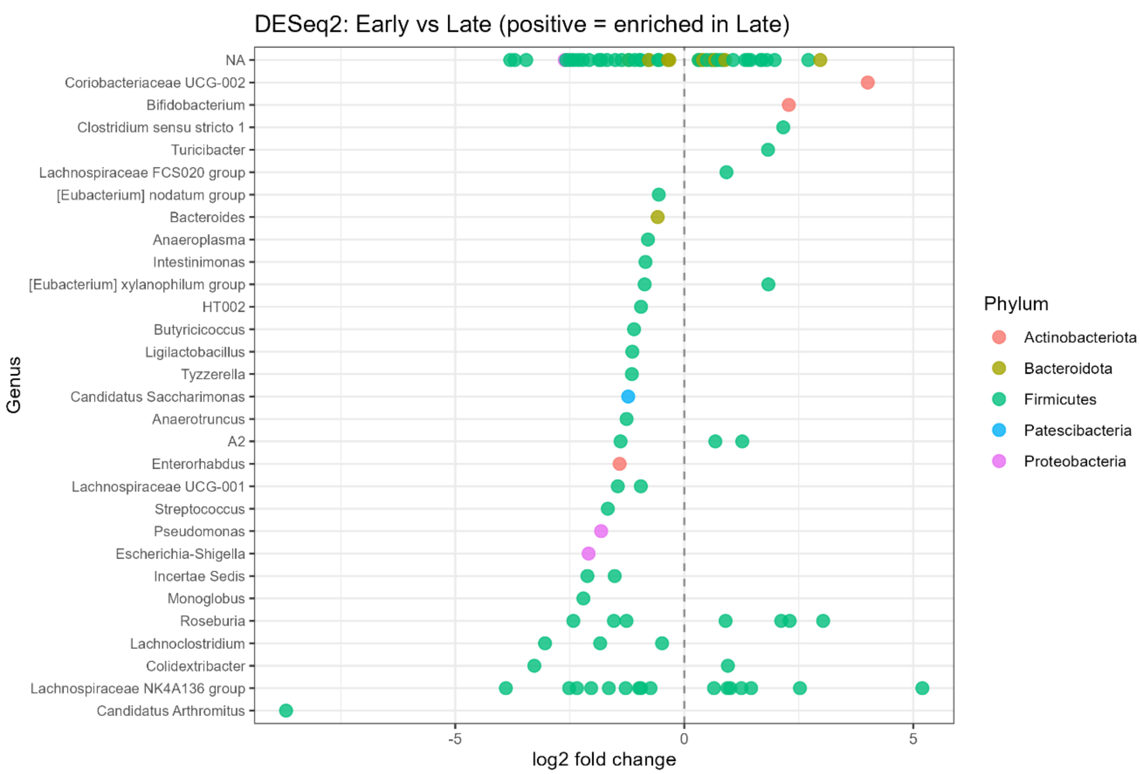

Figure 4. DESeq2 log2 fold-changes for all significant ASVs (padj < 0.05), colored by phylum. Positive values indicate enrichment in Late samples.

ANCOM-BC2 (Genus level)

ANCOM-BC2, run at the genus level with a 10% prevalence filter and bias correction, identified 10 differentially abundant genera between Early and Late samples: 5 depleted in Late (*Candidatus arthromitus*, *Ligilactobacillus*, *Pseudomonas*, *Monoglobus*, *Butyricicoccus*) and 5 enriched in Late (*Bifidobacterium*, *Turicibacter*, *Lactobacillus*, *Clostridium* sensu stricto 1, and A2). As with DESeq2, *Candidatus arthromitus* was the strongest and most significant hit (log-fold-change −3.32, $q = 4.9\times10^{-50}$), reinforcing its role as the dominant driver of the Early/Late shift.

Genus	log-fold-change (Late)	q-value
Candidatus Arthromitus	−3.32	4.9e-50
Bifidobacterium	+0.80	3.8e-07
Turicibacter	+0.83	1.2e-06
Ligilactobacillus	−0.60	1.1e-03
Pseudomonas	−0.51	1.3e-04
Lactobacillus	+0.65	9.3e-04
Clostridium sensu stricto 1	+0.51	9.3e-04
Monoglobus	−0.37	4.4e-03
Butyricicoccus	−0.35	8.0e-03
A2	+0.45	4.4e-03

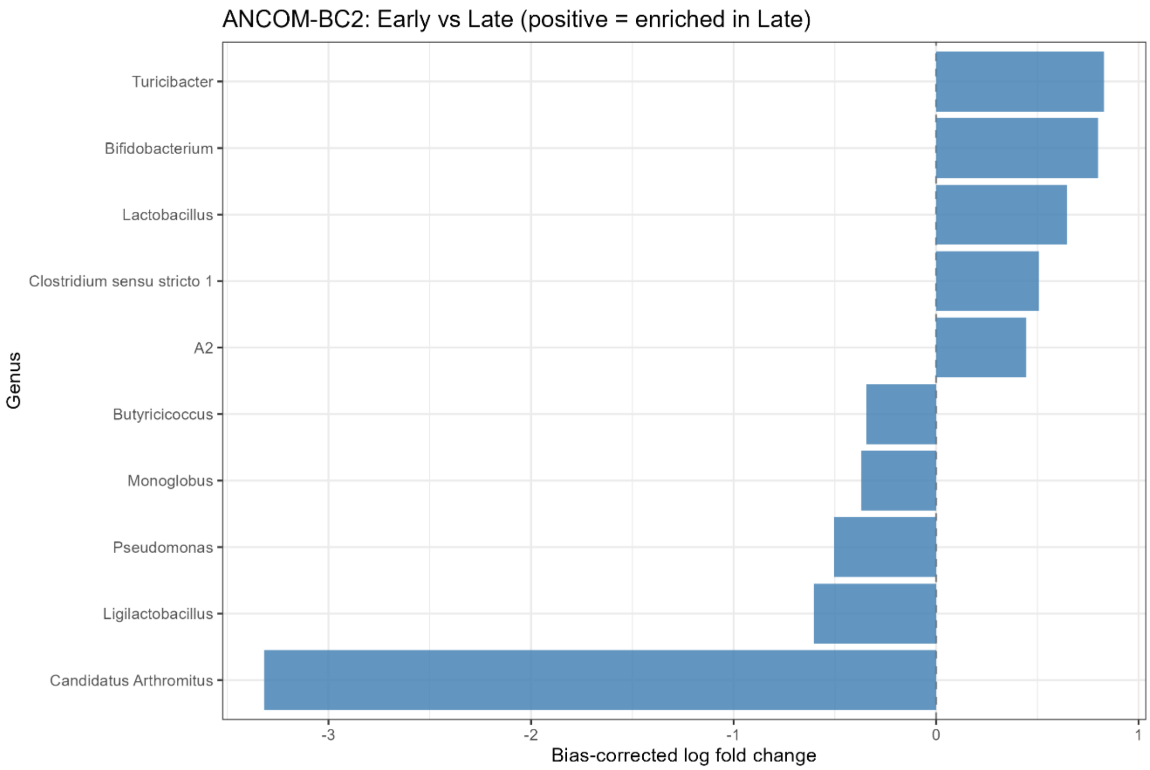

Figure 5. ANCOM-BC2 bias-corrected log-fold-changes for all 10 significant genera. Positive values indicate enrichment in Late samples.

Cross-Method Comparison

DESeq2 flagged 29 unique genera and ANCOM-BC2 flagged 10. 9 of ANCOM-BC2's 10 genera (90%) were also flagged by DESeq2, indicating strong agreement between the two methods for the taxa ANCOM-BC2 considered significant. The 9 overlapping, most-confident genera were: *Candidatus arthromitus*, *Monoglobus*, *Pseudomonas*, A2, *Ligilactobacillus*, *Butyricicoccus*, *Turicibacter*, *Clostridium* sensu stricto 1, and *Bifidobacterium*. DESeq2's larger hit list likely reflects its ASV-level, less conservative testing rather than the presence of many false negatives in ANCOM-BC2; per the interpretation guidance built into the analysis, the 9 overlapping genera should be reported with the highest confidence, and the remaining DESeq2-only genera treated as suggestive but less certain.

Limitations and Data-Quality Notes

Sample-level "Gender" metadata is not usable. The regex used to derive it from the sample ID did not match the naming pattern actually used (all sample IDs begin with "F" followed by a subject number, e.g. F3D0), so the exported Gender field contains the raw sample ID rather than a true sex label. Subject and Day/When fields, which use different, correctly matching patterns, are unaffected and were used as-is throughout this analysis.

The NMDS stress value reported by metaMDS for the Bray-Curtis ordination was unusually low (~0.0002–0.0005). This does not change the qualitative Early/Late separation shown in Figure 2, but the exact stress value should be re-verified before it is quoted as a standalone goodness-of-fit statistic.

Species-level taxonomic resolution was low (5.5% of ASVs), which is expected for the V4 region and should not be interpreted as a processing error.

Two samples lacking an interpretable collection day (likely mock community or control samples) were excluded from the Early/Late diversity and differential abundance analyses.

DESeq2 and ANCOM-BC2 apply prevalence filtering at slightly different points in the workflow (ASV-level vs. internal genus-level), so the exact sets of taxa tested by each method are not identical. This likely explains part of the difference in the number of significant hits between the two methods, independent of any true biological difference in sensitivity.

Conclusion

DADA2 processing of the full 362-sample Schloss mouse-gut dataset produced 420 high-quality, non-chimeric ASVs from 81.4% of the original read pairs, with taxonomic and length-distribution profiles consistent with a well-executed V4-region 16S run. The gut microbiota showed a clear compositional and diversity shift between Early and Late collection windows, and two independent differential-abundance methods (DESeq2 and ANCOM-BC2) converged on a consistent, biologically plausible set of taxa driving that shift, most notably a large, highly significant depletion of *Candidatus Arthromitus* (segmented filamentous bacteria) in Late samples, alongside enrichment of *Bifidobacterium*, *Turicibacter*, and *Lactobacillus*. These results are consistent with the expected trajectory of early-life gut microbiome maturation in mice (Schloss et al., 2012).

References

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